Generated by Scopus AI, Sat Oct 10 2026

Satellite imagery for infrastructure mapping

Satellite Imagery for Rural Infrastructure Mapping: Methodologies, Technologies, and Global Applications

Quick Reference

Key Findings Table

Theme	Summary	Supporting Citations
Advances in Methodologies	Deep learning and entity-based image analysis (EBIA) significantly improve mapping accuracy and scalability; F1 scores 0.79–0.88 for rural settlements, >90% for roads and ecological features	1 2
Multi-Source Data Fusion	Integration of optical, radar, hyperspectral, UAV, and citizen science data enhances spatial/spectral quality and temporal monitoring	3 4 5
Cost-Benefit Trade-offs	Very high resolution imagery offers fine detail but is costly; medium resolution is more accessible but less precise	6 7 8
Data Quality Challenges	Inconsistent quality, lack of metadata, and multi-temporal integration issues impact model performance and transferability	9 10
Local Knowledge Integration	Combining satellite imagery with local knowledge and policy frameworks improves planning and outcomes	11 12 13

Direct Answer

The use of satellite imagery for rural infrastructure mapping has been significantly enhanced by the integration of advanced classification methods such as EBIA and deep learning, along with multi-source data fusion techniques. While very high resolution imagery offers fine detail necessary for mapping small-scale features, cost and data availability challenges make medium resolution imagery more practical for some regions. Integration of local knowledge, GIS, and policy frameworks further enriches planning outcomes, enabling tailored infrastructure interventions across diverse geographic contexts.

Study Scope

- **Time Period:** 2000–2024 (with emphasis on post-2015 advances)
- **Disciplines:** Remote sensing, GIS, machine learning, rural development, environmental science

- **Methods:** Meta-analysis, comparative case studies, empirical evaluation of classification and segmentation models, integration of citizen science and participatory approaches

Assumptions & Limitations

- **Assumptions:** Satellite imagery is available and accessible; ground truth or local knowledge can be integrated; computational resources are sufficient for deep learning models.
- **Limitations:** Data quality and metadata inconsistencies; cost and accessibility of very high resolution imagery; transferability of models across regions and time; underutilization of multi-source data fusion in some contexts; policy and institutional barriers to operational uptake.

Suggested Further Research

- Standardization of metadata and protocols for multi-source, multi-temporal data integration
- Development of scalable, cost-effective mapping methods for resource-constrained regions
- Enhanced participatory and community-driven mapping frameworks
- Integration of advanced sensing (e.g., fiber optic, acoustic) and AI for real-time infrastructure monitoring
- Research on adaptive, context-aware models for dynamic rural infrastructure mapping

1. Introduction

Overview of Satellite Imagery in Rural Infrastructure

Satellite imagery has transformed rural infrastructure mapping, offering unprecedented spatial and temporal coverage for planning, monitoring, and development. The evolution from traditional field surveys to remote sensing has enabled rapid, scalable, and cost-effective mapping of diverse rural features, from homes and roads to ecological buffers and irrigation systems. Recent advances in machine learning, deep learning, and data fusion have further enhanced the precision and utility of satellite-based mapping, supporting sustainable rural development and informed policy-making across the globe [1] [14].

Synthesis: The convergence of remote sensing technologies and computational advances has democratized access to spatial data, enabling both high-level policy interventions and grassroots participatory mapping. However, challenges in data quality, accessibility, and integration remain, particularly in developing regions.

2. Types of Rural Infrastructure Mapped Using Satellite Imagery

Traditional Rural Infrastructure Types

- **Homes and Settlements:** Entity-based image analysis (EBIA) and high-resolution imagery enable accurate mapping of rural homes, settlements, and their spatial footprints, supporting infrastructure planning (e.g.,

electrification) [1] [5].

- **Roads and Agricultural Buildings:** Deep learning models and object-based image analysis delineate road networks, agricultural buildings, and parcels, with high accuracy for paved/unpaved roads and small-scale features [15] [16].

- **Irrigation Systems and Ponds:** Multi-temporal satellite imagery tracks irrigation and drainage systems, supporting water resource management and infrastructure optimization [15] [17].

Emerging and Ecological Infrastructure Types

- **Riparian Buffers and Shelterbelts:** Multispectral and machine learning-based detection identifies ecological features such as riparian buffers, shelterbelts, green spaces, and ecological ditches, with classification accuracies >90% [1] [18] [19] [20].

- **Indigenous Construction Materials:** Remote sensing methods map materials like calcrete for low-cost rural road construction, expanding infrastructure types beyond traditional categories [19].

- **Pedestrian Water Crossings:** Blended approaches combining remote sensing and field surveys improve detection of small-scale infrastructure elements [21].

Role of Ecological Features in Rural Planning

- **Ecosystem Services:** Riparian buffers and shelterbelts provide pollution retention, erosion control, biodiversity support, and water quality improvement. Their effectiveness depends on buffer width, vegetation composition, and management practices [22] [23] [24] [25].

- **Spatial Analysis:** GIS and remote sensing enable spatially explicit planning and prioritization of ecological infrastructure restoration, optimizing benefits for rural communities.

Temporal Monitoring of Infrastructure Changes

- **Multi-temporal Analysis:** Landsat, IRS, and other satellite data track changes in rural infrastructure (e.g., irrigation, roads) over years, using supervised classification and spectral indices (NDVI, SAVI) [26] [27] [28].

- **Predictive Modeling:** Hidden Markov Models and adaptive neural networks forecast land use changes and infrastructure dynamics, supporting long-term planning.

Synthesis: The scope of rural infrastructure mapping has expanded from basic physical features to include ecological and environmental elements, with advanced methodologies enabling both spatial and temporal monitoring. Integration of ecological features into rural planning is increasingly recognized as essential for sustainability.

3. Satellite Imagery Sources and Processing Techniques

Satellite Platforms and Resolution Trade-offs

- **Medium Resolution (Landsat, Sentinel):** Effective for regional mapping; EBIA achieves F1 scores 0.79–0.88 for settlements [1].
- **High/Very High Resolution (Ikonos, PlanetScope):** Enables detailed inventory and characterization; critical for small-scale features but costly [6] [7] [29].
- **Multi-resolution Integration:** Combining CBERS, SPOT, Quickbird, and UAV data improves extraction accuracy [30] [31].

Platform	Spatial Resolution	Typical Use Cases	Cost/Accessibility
Landsat	30m	Regional land cover, change	High accessibility
Sentinel-2	10m	Vegetation, infrastructure	High accessibility
Ikonos	1m	Building, road mapping	Moderate cost
PlanetScope	3–5m	Parcel-level mapping	Higher cost
UAV Orthomosaics	<0.1m	Cadastral, small features	High cost, local

Image Processing and Semantic Segmentation Techniques

- **Deep Learning Models:** Hybrid DenseNet-U-Net, DeepLabV3+, transformer-based models achieve state-of-the-art segmentation, with Dice coefficients up to 91.3% [2] [32] [33] [34].
- **Attention Mechanisms:** Improve boundary detection and feature extraction, especially for complex rural features [35] [36].
- **Object-Based and Entity-Based Analysis:** OBIA and EBIA outperform pixel-based methods, especially for heterogeneous rural landscapes [1] [32].

Multi-source Data Fusion and Quality Enhancement

- **Hyperspectral, Radar, Optical Integration:** Multi-platform data acquisition balances spatial/spectral resolution, cost, and coverage, improving infrastructure extraction [3] [37].
- **Data Fusion Techniques:** Enhance spatial and spectral quality, reduce cloud cover limitations, and enable frequent updates [4].

Citizen Science and Crowdsourced Annotations

- **East Africa Case Studies:** Citizen science annotations combined with very high resolution imagery produce large-scale rural home datasets (recall 93%, precision 49–69%), supporting energy access planning [5] [38] [39].

- **Integration with Machine Learning:** Enhances spatial mapping accuracy and contextual understanding, especially in data-poor regions.

Synthesis: Advances in image processing, semantic segmentation, and data fusion have enabled more precise and scalable rural infrastructure mapping. Cost-benefit trade-offs and accessibility issues remain, particularly for very high resolution imagery in developing countries. Citizen science initiatives are increasingly important for supplementing satellite data and improving mapping outcomes.

4. Machine Learning and Deep Learning Approaches

Classification and Segmentation Model Performance

- **EBIA:** Groups related land cover elements, achieving high F1 scores (0.79–0.88) and outperforming traditional methods [1].

- **Deep Learning (CNNs, U-Net, DeepLabV3+):** Superior accuracy for building and road extraction; recall/precision for unpaved roads 89–97%/85–91% [33] [40] [41].

- **Hybrid Models:** DenseNet-U-Net and transformer-based architectures improve feature extraction and boundary detection [34].

Handling Class Imbalance and Computational Costs

- **Class Imbalance:** Data augmentation, focal loss, and GAN-based methods mitigate imbalance; critical for rural datasets with skewed distributions [42].

- **Computational Efficiency:** Spatial partitioning and lightweight architectures (e.g., MobileNetV2) reduce costs and improve scalability [43] [44].

Domain Adaptation and Fairness Across Regions

- **Adversarial Learning, Mixup, Contrastive Learning:** Improve model fairness and transferability across heterogeneous regions; domain adaptation methods achieve significant IoU gains [45] [46] [47] [48].

- **Open-set and Domain-agnostic Adaptation:** Address label set discrepancies and generalizability without explicit domain labels.

Temporal Variability and Phenology in Model Transferability

- **Phenological Variability:** Reduces model transferability; normalization (e.g., growing degree days) and multi-year training improve robustness [9] [49] [50].

- **Adaptive Modeling:** Continuous updating and retraining recommended for dynamic rural landscapes.

Synthesis: Machine learning and deep learning models have revolutionized rural infrastructure mapping, but challenges in class imbalance, computational costs, and domain adaptation persist. Temporal variability in vegetation phenology is a major barrier to model transferability, requiring normalization and adaptive strategies.

5. Challenges and Limitations in Rural Infrastructure Mapping

Technical and Data Quality Challenges

- **Image Quality and Metadata:** Varying quality, imbalances between urban/rural data, and lack of metadata impede analysis and model performance [9] [10] [51].
- **Georeferencing Accuracy:** Airborne and UAV images often provide better precision than some satellite images [52].
- **Underutilization:** Lack of knowledge about satellite image properties leads to underexploitation in rural mapping [53].

Integration of Multi-temporal and Multi-resolution Data

- **Pre-processing and Calibration:** Spatial/temporal inconsistencies require geometric/radiometric correction and calibration across sensors [10] [54] [55].
- **Data Standards and Infrastructure:** Fragmented standards and insufficient infrastructure limit comprehensive monitoring [56].
- **Physical Distortions:** Atmospheric, solar, and sensor variations complicate model portability and data integration [57].

Interpretation Limitations and Policy Implications

- **Biases and Disparities:** Urban-rural disparities and prediction errors in satellite-based poverty maps can undermine policy decisions [58].
- **Context-specific Approaches:** Limitations in interpretation necessitate integration with local knowledge and participatory methods for effective policy-making [13] [59].

Synthesis: Technical, practical, and policy-related challenges remain significant barriers to robust rural infrastructure mapping. Data quality, metadata, and integration issues impact both model performance and the reliability of policy decisions, underscoring the need for context-specific and participatory approaches.

6. Case Studies and Geographic Variations

Regional Applications and Methodological Differences

- **Africa:** Focus on infrastructure challenges (road type, electrification) using high-resolution imagery and deep learning; data fusion to overcome cloud cover and cost limitations [60] [61].

- **India:** Satellite imagery supports irrigation, energy access, and participatory planning; integration with local knowledge reveals socio-institutional vulnerabilities [11] [12].

- **Europe:** Multi-source data integration (Sentinel, UAV, IoT) for precision agriculture; policy frameworks (CAP, RDP) and open-source tools support comprehensive monitoring [13] [62] [63] [64].

Citizen Science and Rural Electrification in East Africa

- **Home Annotation Datasets:** Citizen science initiatives produce large datasets for electrical infrastructure planning; machine learning integration improves mapping precision [5] [39] [65].

Integrating Satellite Data with Local Knowledge

- **Participatory Approaches:** Combining satellite imagery with local knowledge and GIS enables targeted interventions and operationalizes green infrastructure planning [12] [66] [67] [68].

Satellite Data and Irrigation Infrastructure in the Senegal River Valley

- **Long-term Monitoring:** Satellite data over 30 years shows increased cultivation rates and climate resilience, but persistent water access constraints highlight the need for integrated management [69] [70] [71] [72].

Socio-institutional Challenges in Rural India

- **Infrastructure Sustainability:** Satellite-based land use analysis reveals deficits in irrigation, roads, credit, and governance, affecting sustainability and resilience [73] [74] [75] [76].

European Precision Agriculture and Policy Integration

- **Multi-source Data and Policy:** Integration of satellite, UAV, IoT, and VGI data within SDI and CAP frameworks supports precision agriculture and infrastructure monitoring; open-source tools and standardization efforts facilitate data sharing [13] [62] [63] [64].

Synthesis: Case studies across Africa, India, and Europe illustrate geographic variations in satellite imagery applications, methodologies, and outcomes. Integration of local knowledge, participatory approaches, and policy frameworks is critical for effective rural infrastructure planning and sustainability.

7. Future Directions and Research Gaps

Advanced Sensing and AI Integration

- **Distributed Fiber Optic Sensing, Acoustic Evaluation:** Emerging technologies offer potential for real-time infrastructure monitoring and expanded mapping capabilities [20] [77] [78].

- **AI Integration:** Advanced models and hybrid architectures improve accuracy and scalability, but computational complexity and data requirements remain challenges.

Participatory and Community-driven Mapping

- **Citizen Science and Local Knowledge:** Participatory methods and integration with satellite imagery enhance sustainability and context sensitivity; operational uptake requires addressing technical and institutional barriers [38] [66] [79].

Data Fusion, Temporal Dynamics, and Scalability

- **Standardization and Integration:** Research gaps in metadata standardization, multi-source data fusion, and temporal dynamics limit scalability and operational use, especially in developing regions [43] [56] [80].
- **Adaptive, Context-aware Models:** Development of dynamic, community-informed mapping systems leveraging real-time satellite data, mobile platforms, and IoT devices is an emerging opportunity.

Synthesis: The field is poised for further innovation, with advanced sensing, AI integration, and participatory approaches offering pathways to more effective and sustainable rural infrastructure mapping. Addressing challenges in data quality, integration, and scalability will be critical for operational uptake and impact.

Conclusion

Satellite imagery has become an indispensable tool for rural infrastructure mapping, enabling detailed, scalable, and cost-effective monitoring and planning across diverse contexts. Advances in machine learning, deep learning, and data fusion have significantly improved mapping accuracy and utility, while participatory approaches and integration with local knowledge enhance sustainability and relevance. Persistent challenges in data quality, metadata, and integration underscore the need for further research and innovation, particularly in developing regions. The future of rural infrastructure mapping lies in adaptive, context-aware systems that harness both technological advances and community engagement for effective rural development.

References

1. Entity-based image analysis: A new strategy to map rural settlements from Landsat images Wang, Y., Zhu, X., Wei, T., (...), Zhang, H. Remote Sensing of Environment, 2025 https://www.scopus.com/pages/publications/85211968307?origin=scopusAI

2. Enhancing land use and land cover classification with deep learning-based satellite imagery segmentation Deressu, T.F., Bojer, A.K., Debelee, T.G., (...), Gebissa, K.W. International Journal of Applied Earth Observation and Geoinformation, 2025 https://www.scopus.com/pages/publications/105016583281?origin=scopusAI

3. Hyperspectral imaging utility for transportation systems Bridgelall, R., Rafert, J.B., Tolliver, D. Proceedings of SPIE - The International Society for Optical Engineering, 2015 https://www.scopus.com/pages/publications/84943403682?origin=scopusAI

4. Advancing spatio-temporal fusion with spectral super-resolution for enhancing remote sensing of marine debris and landfills Kremezi, M., Karathanassi, V., Taggio, N., (...), Ceriola, G. European Journal of Remote Sensing, 2026 https://www.scopus.com/pages/publications/105051470853?origin=scopusAI

5. Rural Home Annotation Dataset Mapped by Citizen Scientists in Satellite Imagery Leonard, A., Wheeler, S., McCulloch, M. Data in Brief, 2022 https://www.scopus.com/pages/publications/85130614992?origin=scopusAI

6. Satellite mapping in Bhutan: High-resolution imagery in generating spatial information Fraser, C., Tshering, D., Gruen, A. GIM International, 2008 https://www.scopus.com/pages/publications/42649120951?origin=scopusAI

7. Perspective of Chinese GF-1 high-resolution satellite data in agricultural remote sensing monitoring ZHOU, Q.-B., YU, Q.-Y., LIU, J., (...), TANG, H.-J. Journal of Integrative Agriculture, 2017 https://www.scopus.com/pages/publications/85012280194?origin=scopusAI

8. Implementation of mixed communication solutions, satellite and Wi-Fi, applied to agriculture experiments conducted in the french agricultural sector within the framework of the european project Twister Boffety, D., Barbe, E., Chanet, J.-P., (...), Vigier, F. Computers in Agriculture and Natural Resources - Proceedings of the 4th World Congress, 2006 https://www.scopus.com/pages/publications/58249119547?origin=scopusAI

9. Transferability of Machine Learning Models for Crop Classification in Remote Sensing Imagery Using a New Test Methodology: A Study on Phenological, Temporal, and Spatial Influences Hoppe, H., Dietrich, P., Marzahn, P., (...), Borg, E. Remote Sensing, 2024 https://www.scopus.com/pages/publications/85193018680?origin=scopusAI

10. Taking the Long View: Enhancing Learning On Multi-Temporal, High-Resolution, and Disparate Remote Sensing Data Correa, S., Perez, G., Jaramillo, P., Taneja, J. BuildSys 2023 - Proceedings of the10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2023 https://www.scopus.com/pages/publications/85179504298?origin=scopusAI

11. The essence of geographic information systems (GIS) in sustainable tourism, public leadership and inclusive community participation in remote-African rural societies Ramaano, A.I. Journal of Responsible Production and Consumption, 2025 https://www.scopus.com/pages/publications/105024709809?origin=scopusAI

12. GIS for Rural Infrastructure Mapping: Aurangpur Case Study Ramesh Babu, C., Kumar, A. SSRG International Journal of Civil Engineering, 2026 https://www.scopus.com/pages/publications/105030728162?origin=scopusAI

13. The Integration of Satellite Analysis Within Common Agricultural Policy Instruments: Application Hypotheses for the Issue of Soil Erosion in the Veneto Region Bonora, A., Lucertini, G., Balletti, C. Communications in Computer and Information Science, 2025 https://www.scopus.com/pages/publications/105018662111?origin=scopusAI

14. Targeting Villages for Rural Development Using Satellite Image Analysis Varshney, K.R., Chen, G.H., Abelson, B., (...), Spatocco, B.L. Big Data, 2015 https://www.scopus.com/pages/publications/84991783905?origin=scopusAI

15. Irrigation and drainage system as-built map preparation using satellite digital imagery and a GIS Neale, Christopher M.U., Willardson, Lymann S. Irrigation and Drainage: Saving a Threatened Resource - In Search of Solutions, Proceedings of the Irrigation and Drainage Sessions at Water Forum '92, 1992 https://www.scopus.com/pages/publications/0027067429?origin=scopusAI

16. The implementation of satellite images and associated digital image processing in addition to GIS modelling for urban mapping in Amman area, Jordan Fadda, E.H.R., Kakish, M., Al Azab, T.A. WSEAS Transactions on Communications, 2009 https://www.scopus.com/pages/publications/66349136183?origin=scopusAI

17. Land cover mapping using Sentinel-2 images in area dominated by small-scale, heterogeneous agricultural parcels Indarto, I., Rahayu, C.U., Soekarno, S. Agriculture and Natural Resources, 2022 https://www.scopus.com/pages/publications/85130705375?origin=scopusAI

18. Evaluation of the ecological performance of rural green infrastructure: a multidimensional assessment framework integrating remote sensing ecological indices and spatial connectivity Ju, H., Liu, Y. GeoJournal , 2026 https://www.scopus.com/pages/publications/105048619420?origin=scopusAI

19. Remote sensing for the location and mapping of engineering construction materials in developing countries Beaumont, T.E. Quarterly Journal of Engineering Geology, 1979 https://www.scopus.com/pages/publications/0004443379?origin=scopusAI

20. Fusion of Distributed Fiber Optic Sensing, Acoustic NDE, and Artificial Intelligence for Infrastructure Monitoring Ohodnicki, P.R., Zhang, P., Lalam, N., (...), Wright, R. Optics InfoBase Conference Papers, 2022 https://www.scopus.com/pages/publications/85146664710?origin=scopusAI

21. Identifying water crossings in rural liberia and rwanda using remote and field-based methods Shirley, K., Noriega, A., Levin, D., Barstow, C. Sustainability (Switzerland), 2021 https://www.scopus.com/pages/publications/85099142087?origin=scopusAI

22. Combining ecosystem service value and landscape ecological risk to subdivide the riparian buffer zone of the Weihe River in Shaanxi Xie, Y., Zhu, Q., Bai, H., (...), Zhang, Y. Ecological Indicators, 2024 https://www.scopus.com/pages/publications/85200500095?origin=scopusAI

23. Understanding Stakeholder Perspectives on the Implementation and Management of Riparian Buffer Zones in the Santa Lucía River Basin, Uruguay Paarlberg, A., Sena, G., Loc, H.H., Schultner, J. Environmental Management, 2025 https://www.scopus.com/pages/publications/105011707957?origin=scopusAI

24. Applications of Remote Sensing and GIS to the Assessment of Riparian Zones for Environmental Restoration in Agricultural Watersheds Gu, F., Liu, W. Geo-Spatial Information Science, 2010 https://www.scopus.com/pages/publications/78449236447?origin=scopusAI

25. Importance of forest buffers for preserving soil carbon and nutrient stocks in farmed landscapes along two river sites in the savannas of the Volta basin, Ghana Boakye, E.A., Hyppolite, D.N., Barnes, V.R., Porembski, S. Arid Land Research and Management, 2017 https://www.scopus.com/pages/publications/85011265878?origin=scopusAI

26. Temporal Variation of Land use in Al-Husainiya Project, Karbala Governorate - Iraq using some Spectral Index Mahdi, N.A., Ali, A.K. IOP Conference Series: Earth and Environmental Science, 2023 https://www.scopus.com/pages/publications/85181576440?origin=scopusAI

27. Contribution of satellite imagery to optimization of the farm irrigation in the Senegal middle valley Seguis, L., Chevillotte, H., Gascuel, C., (...), Braudeau, E. Teledetection appliquee a la cartographie thematique et topographique. Actes des journees scientifiques, Montreal, 1991, 1993 https://www.scopus.com/pages/publications/0027708617?origin=scopusAI

28. Selected farming services with satellite assistance: Mapping crops, optimizing water resources and assessing crop and pasture drought stress Calera, A., Lozano, F.J., Campos, I., Osann, A. American Society for Photogrammetry and Remote Sensing Annual Conference, ASPRS 2013, 2013 https://www.scopus.com/pages/publications/84887910041?origin=scopusAI

29. The application of new technologies in construction: Inventory and characterisation of rural constructions using the Ikonos satellite image Armesto González, J., Gil Docampo, M.L., Cañas Guerrero, I. Building and Environment, 2006 https://www.scopus.com/pages/publications/26444460753?origin=scopusAI

30. Comparison of extracting rural residential area from satellite images with multiresolution Yang, C., Wang, X., Huang, H. International Geoscience and Remote Sensing Symposium (IGARSS), 2008 https://www.scopus.com/pages/publications/67649806877?origin=scopusAI

31. Semantic Segmentation of High-Resolution Remote Sensing Images Based on RS³Mamba: An Investigation of the Extraction Algorithm for Rural Compound Utilization Status Fang, X., Liu, Z., Xie, S., Ge, Y. Remote Sensing, 2025 https://www.scopus.com/pages/publications/105020251756?origin=scopusAI

32. Deep hybrid network for land cover semantic segmentation in high-spatial resolution satellite images Khan, S.D., Alarabi, L., Basalamah, S. Information (Switzerland), 2021 https://www.scopus.com/pages/publications/85107833239?origin=scopusAI

33. Evaluation of Deep Learning Models for Remote Sensing Segmentation and Classification Ananth, S., Gangashetty, S. International Arab Journal of Information Technology, 2025 https://www.scopus.com/pages/publications/105022900880?origin=scopusAI

34. An In-Depth Analysis of U-Net Variations for Advancing Precision in Medical Image Segmentation Athira, R., Prakash, P. 10th International Conference on Advanced Computing and Communication Systems, ICACCS 2024, 2024 https://www.scopus.com/pages/publications/85208624676?origin=scopusAI

35. MSCSA-Net: Multi-Scale Channel Spatial Attention Network for Semantic Segmentation of Remote Sensing Images Liu, K.-H., Lin, B.-Y. Applied Sciences (Switzerland), 2023 https://www.scopus.com/pages/publications/85170404728?origin=scopusAI

36. Empowering rural governance with digital technology: Deep learning models for automated detection of rural buildings using remote sensing images Zhong, J., Xie, Y., Li, L., Shi, C. PLOS ONE, 2026 https://www.scopus.com/pages/publications/105042364766?origin=scopusAI

37. Current trends in urban remote sensing Gamba, P. 30th Asian Conference on Remote Sensing 2009, ACRS 2009, 2009 https://www.scopus.com/pages/publications/84866096818?origin=scopusAI

38. Tracking Rangeland Degradation from Prosopis juliflora Invasion in Kenya: A Multi-Source Approach Combining Remote Sensing, Machine Learning, and Citizen Science Cherotich, F., Galgallo, D., Dhulipala, R., (...), Paliwal, A. Ecologies, 2026 https://www.scopus.com/pages/publications/105034290255?origin=scopusAI

39. Power to the people: Applying citizen science and computer vision to home mapping for rural energy access Leonard, A., Wheeler, S., McCulloch, M. International Journal of Applied Earth Observation and Geoinformation, 2022 https://www.scopus.com/pages/publications/85129354692?origin=scopusAI

40. Maintaining accurate, current, rural road network data: An extraction and updating routine using RapidEye, participatory GIS and deep learning Kearney, S.P., Coops, N.C., Sethi, S., Stenhouse, G.B. International Journal of Applied Earth Observation and Geoinformation, 2020 https://www.scopus.com/pages/publications/85086574683?origin=scopusAI

41. Scene Segmentation in rural and urban regions from Remote Sensing data Sivaprakasam, S.A., Prasanth, K., Kishore, D.R. Proceedings of 2026 1st International Conference on Intelligent Computing and Automation for Sustainable Solutions, ICASS 2026, 2026 https://www.scopus.com/pages/publications/105042853459?origin=scopusAI

42. Addressing class imbalance in remote sensing using deep learning approaches: a systematic literature review Sharma, S., Gosain, A. Evolutionary Intelligence, 2025 https://www.scopus.com/pages/publications/85217717656?origin=scopusAI

43. Small is beautiful: Distributed orchestration of spatial deep learning workloads Rammer, D., Bruhwiler, K., Khandelwal, P., (...), Lee Pallickara, S. Proceedings - 2020 IEEE/ACM 13th International Conference on Utility and Cloud Computing, UCC 2020, 2020 https://www.scopus.com/pages/publications/85099597948?origin=scopusAI

44. MobileNetV2-Based Framework for Scalable and Accurate Remote Sensing Image Classification Tuteja, G., Manchanda, R., Badhani, P., (...), Gupta, R. Proceedings of the International Conference on Electrical, Electronics, and Computer Science with Advance Power Technologies - A Future Trends, ICE2CPT 2025, 2025 https://www.scopus.com/pages/publications/105033360548?origin=scopusAI

45. Weakly-supervised domain adaptation for built-up region segmentation in aerial and satellite imagery Iqbal, J., Ali, M. ISPRS Journal of Photogrammetry and Remote Sensing, 2020 https://www.scopus.com/pages/publications/85089375492?origin=scopusAI

46. Cross-Domain Mixup for Parcel-Level Crop Mapping on a Multi-Year Sentinel-2 Dataset from Slovakia Dinescu, A.-A., Florea, C. Remote Sensing, 2026 https://www.scopus.com/pages/publications/105050226662?origin=scopusAI

47. Mitigating Urban-Rural Disparities in Contrastive Representation Learning with Satellite Imagery Zhang, M., Chunara, R. Proceedings of the 7th AAAI/ACM Conference on AI, Ethics, and Society, AIES 2024, 2024 https://www.scopus.com/pages/publications/105040250167?origin=scopusAI

48. Domain-Agnostic Domain Adaption for Building Footprint Extraction Zhang, F., Shi, Y., Zhu, X.X. International Geoscience and Remote Sensing Symposium (IGARSS) , 2022 https://www.scopus.com/pages/publications/85140392212?origin=scopusAI

49. The yield strikes back: Enhancing the transferability of field scale wheat and barley yield models by leveraging Sentinel-1/2 Franch, B., Moletto-Lobos, I., Tarín-Mestre, J., (...), Cantero-Martinez, C. International Journal of Applied Earth Observation and Geoinformation, 2026 https://www.scopus.com/pages/publications/105029128227?origin=scopusAI

50. Temporal transferability of wildlife habitat models: Implications for habitat monitoring Tuanmu, M.-N., Viña, A., Roloff, G.J., (...), Liu, J. Journal of Biogeography, 2011 https://www.scopus.com/pages/publications/79960262941?origin=scopusAI

51. Understanding economic development in rural Africa using satellite imagery, building footprints and deep models Elmustafa, A., Rozi, E., He, Y., (...), Lobell, D. GIS: Proceedings of the ACM International Symposium on Advances in Geographic Information Systems, 2022 https://www.scopus.com/pages/publications/85143607022?origin=scopusAI

52. Evaluation of the positional accuracy of features obtained by images of orbital sensors and airborne for georeferencing of rural properties de Paiva, A.A., Burim, S.C., Borges, P.A.F., Lacerda, C.S.A. Revista Brasileira de Geografia Fisica, 2021 https://www.scopus.com/pages/publications/85125648447?origin=scopusAI

53. A review on availability of remote sensing data Jenice Aroma, R., Raimond, K. Proceedings - 2015 IEEE International Conference on Technological Innovations in ICT for Agriculture and Rural Development, TIAR 2015, 2015 https://www.scopus.com/pages/publications/84988976280?origin=scopusAI

54. The use of the digital terrain model for land use mapping in the Ardeche area of France Hill, J., Kohl, H., Megier, J. Proceedings of the ninth EARSeL symposium, Espoo, 1989, 1990 https://www.scopus.com/pages/publications/0025621411?origin=scopusAI

55. Empirical cross sensor comparison of Sentinel-2A and 2B MSI, Landsat-8 OLI, and Landsat-7 ETM+ top of atmosphere spectral characteristics over the conterminous United States Chastain, R., Housman, I., Goldstein, J., Finco, M. Remote Sensing of Environment, 2019 https://www.scopus.com/pages/publications/85057001016?origin=scopusAI

56. Progress and prospect of spatio-temporal data pedestal for Three Rural Issues of modernization in China Cao, X., Chi, S., Liu, Z., (...), Shi, S. Discover Applied Sciences, 2026 https://www.scopus.com/pages/publications/105051176672?origin=scopusAI

57. Multispectral land-use/land-cover model portability in multi-temporal multi-angle very high resolution imagery Longbotham, N., Emery, W., Pacifici, F. International Geoscience and Remote Sensing Symposium (IGARSS), 2013 https://www.scopus.com/pages/publications/84894279876?origin=scopusAI

58. Fairness and Representation in Satellite-Based Poverty Maps: Evidence of Urban-Rural Disparities and Their Impacts on Downstream Policy Aiken, E., Rolf, E., Blumenstock, J. IJCAI International Joint Conference on Artificial Intelligence, 2023 https://www.scopus.com/pages/publications/85170399802?origin=scopusAI

59. Fitness of Multi-Resolution Remotely Sensed Data for Cadastral Mapping in Ekiti State, Nigeria Taiwo, I.O., Ibitoye, M.O., Oladejo, S.O., Koeva, M. Remote Sensing, 2024 https://www.scopus.com/pages/publications/85206483255?origin=scopusAI

60. Identification of road types from satellite images Badji, P., Carroccia, R., Rocco, M.D., (...), Usami, D.S. 2025 3rd International Conference on Artificial Intelligence and Automation Control, AIAC 2025, 2025 https://www.scopus.com/pages/publications/105032641717?origin=scopusAI

61. Crop type and land cover mapping in northern malawi using the integration of sentinel-1, sentinel-2, and planetscope satellite data Kpienbaareh, D., Sun, X., Wang, J., (...), Dakishoni, L. Remote Sensing, 2021 https://www.scopus.com/pages/publications/85100934021?origin=scopusAI

62. Big Data from Space for Precision Agriculture Applications Bovolo, F., Bruzzone, L., Fernández-Prieto, D., (...), Zanetti, M. IOP Conference Series: Earth and Environmental Science, 2020 https://www.scopus.com/pages/publications/85088503515?origin=scopusAI

63. FARMSTAR: An efficient decision support tool for near real time crop management from satellite images Coquil, B., Bordes, J.P. Precision Agriculture 2005, ECPA 2005, 2005 https://www.scopus.com/pages/publications/84893399339?origin=scopusAI

64. Standardisation in precision agriculture through INSPIRE Korduan, P., Bill, R. Precision Agriculture 2013 - Papers Presented at the 9th European Conference on Precision Agriculture, ECPA 2013, 2013 https://www.scopus.com/pages/publications/84893421252?origin=scopusAI

65. PowerScour: Tracking Electrified Settlements Using Satellite Data Correa, S., Shah, Z., Wu, Y., (...), Taneja, J. BuildSys 2022 - Proceedings of the 2022 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2022 https://www.scopus.com/pages/publications/85144615052?origin=scopusAI

66. Moving green infrastructure planning from theory to practice in sub-Saharan African cities requires collaborative operationalization Breed, C.A., Du Plessis, T., Engemann, K., (...), Pasgaard, M. Urban Forestry and Urban Greening, 2023 https://www.scopus.com/pages/publications/85173182695?origin=scopusAI

67. Rural empowerment thru' space technology enabled Village Resource Centres (VRCs) Hegde, V.S., Jayaraman, V., Bhaskaranarayana, A., (...), Ghosh, R. International Astronautical Federation - 58th International Astronautical Congress 2007, 2007 https://www.scopus.com/pages/publications/55649115988?origin=scopusAI

68. Indian scenario in the application of Geographical Information System and geo-informatics Narayan, L.R.A. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives, 2000 https://www.scopus.com/pages/publications/85046355780?origin=scopusAI

69. Irrigation infrastructure and satellite-measured land cultivation impacts: Evidence from the Senegal river valley Cisse, A., de Janvry, A., Ferguson, J., (...), Syll, M.M.A. Journal of Development Economics, 2026 https://www.scopus.com/pages/publications/105015947964?origin=scopusAI

70. Assessing climate resilience in rice production: measuring the impact of the Millennium Challenge Corporation’s IWRM scheme in the Senegal River Valley using remote sensing and machine learning Fionnagáin, D.I., Geever, M., O’Farrell, J., (...), Golden, A. Environmental Research Letters, 2024 https://www.scopus.com/pages/publications/85198633579?origin=scopusAI

71. Remote sensing in support for monitoring rice in the Senegal catchment Lefrancq, M., Diarra, M., Hane, O., (...), Yena, M. Houille Blanche, 2020 https://www.scopus.com/pages/publications/85097634347?origin=scopusAI

72. Rapid expansion of irrigated agriculture in the Senegal River Valley following the 2008 food price crisis Higginbottom, T.P., Adhikari, R., Foster, T. Environmental Research Letters, 2023 https://www.scopus.com/pages/publications/85146487661?origin=scopusAI

73. Integrating Land Use and Community Perspectives for Sustainable Agriculture in Rural India, Dewgain Ejigu, D., Pushpalatha, R., Sajithkumar, K.J., (...), Himanshu, S.K. Sustainable Development, 2026 https://www.scopus.com/pages/publications/105039679963?origin=scopusAI

74. Rural infrastructure, agricultural development and poverty in India: An inter-state study Singh, R. Journal of Rural Development, 2004 https://www.scopus.com/pages/publications/3543070567?origin=scopusAI

75. Leveraging MPLADS Funds as a Catalyst for Rural Infrastructure Enhancement in India Thadaboina, V., Raja, G.S.R.K. Journal of Rural Development, 2025 https://www.scopus.com/pages/publications/105015405217?origin=scopusAI

76. Impact of Land Use Change on Livelihood Options: A Case Study of Upper Pasolgad Watershed, Uttarakhand Pandey, B.W., Prasad, A.S., Mahto, J.K. Sustainable Development Goals Series, 2021 https://www.scopus.com/pages/publications/85195120246?origin=scopusAI

77. Analysis of Modern Landscape Architecture Evolution Using Image-Based Computational Methods Zhang, J., Gao, C. Mathematics, 2025 https://www.scopus.com/pages/publications/105015366912?origin=scopusAI

78. Cross-Spatial Pixel Integration and Cross-Stage Feature Fusion-Based Transformer Network for Remote Sensing Image Super-Resolution Lu, Y., Min, L., Wang, B., (...), Long, T. IEEE Transactions on Geoscience and Remote Sensing, 2023 https://www.scopus.com/pages/publications/85178067683?origin=scopusAI

79. Empowering the rural poor through eo products and services - An impact assessment Diwakar, P.G., Ranganath, B.K., Gowrisankar, D., Jayaraman, V. International Astronautical Federation - 58th International Astronautical Congress 2007, 2007 https://www.scopus.com/pages/publications/55649091416?origin=scopusAI

80. Leveraging Remote Sensing Technologies: A Review of Precision Cropping Challenges and Research Trend Analysis Kuunya, R., Mohammed, E.A.H., Osman, M.M.A., (...), Ragán, P. Journal of the Indian Society of Remote Sensing, 2026 https://www.scopus.com/pages/publications/105040247567?origin=scopusAI